

Tecnológico de Monterrey

Instituto Tecnológico de Estudios Superiores de Monterrey

Campus Toluca

Interconexión de dispositivos (Gpo 301)

10.4.3 Laboratorio - Armar una red con un switch y un router

Profesores:

Yerly Flores Garcia

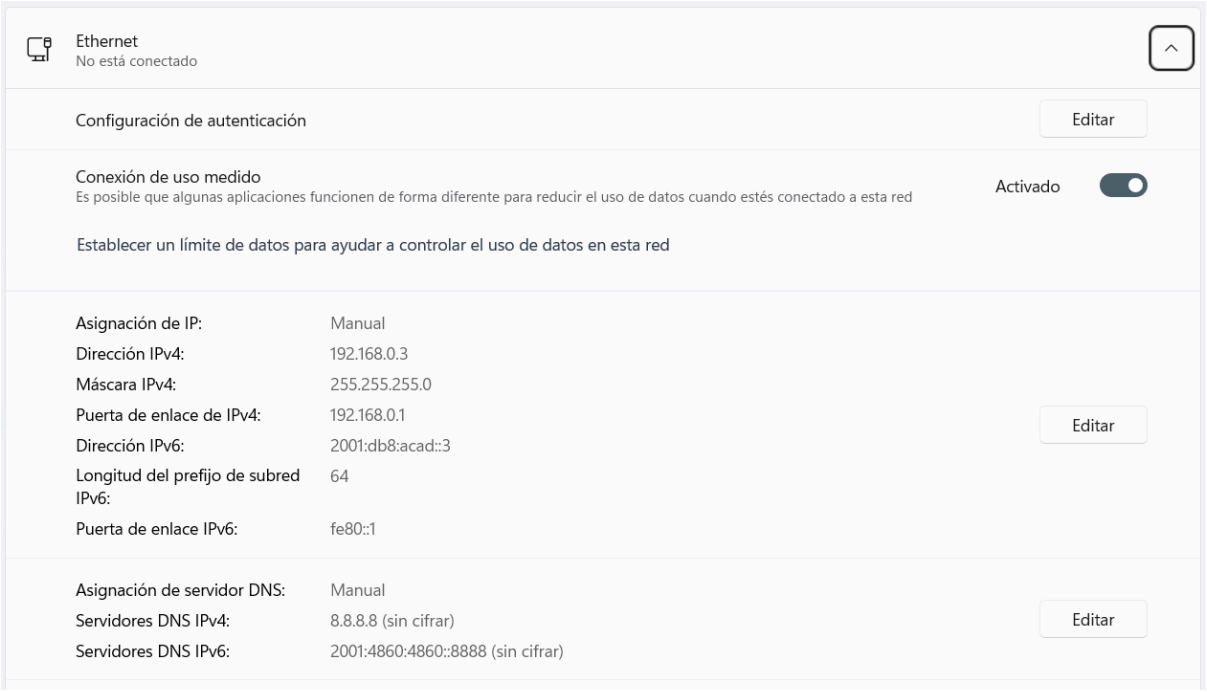
José María Aguilera Méndez

Alan Joseph Salazar Segura - A01773491

Ileana Tapia Castillo - A01773374

Fátima M. Herrera González - A01770839

Parte 3, paso 1



¿Por qué los pings no fueron correctos?

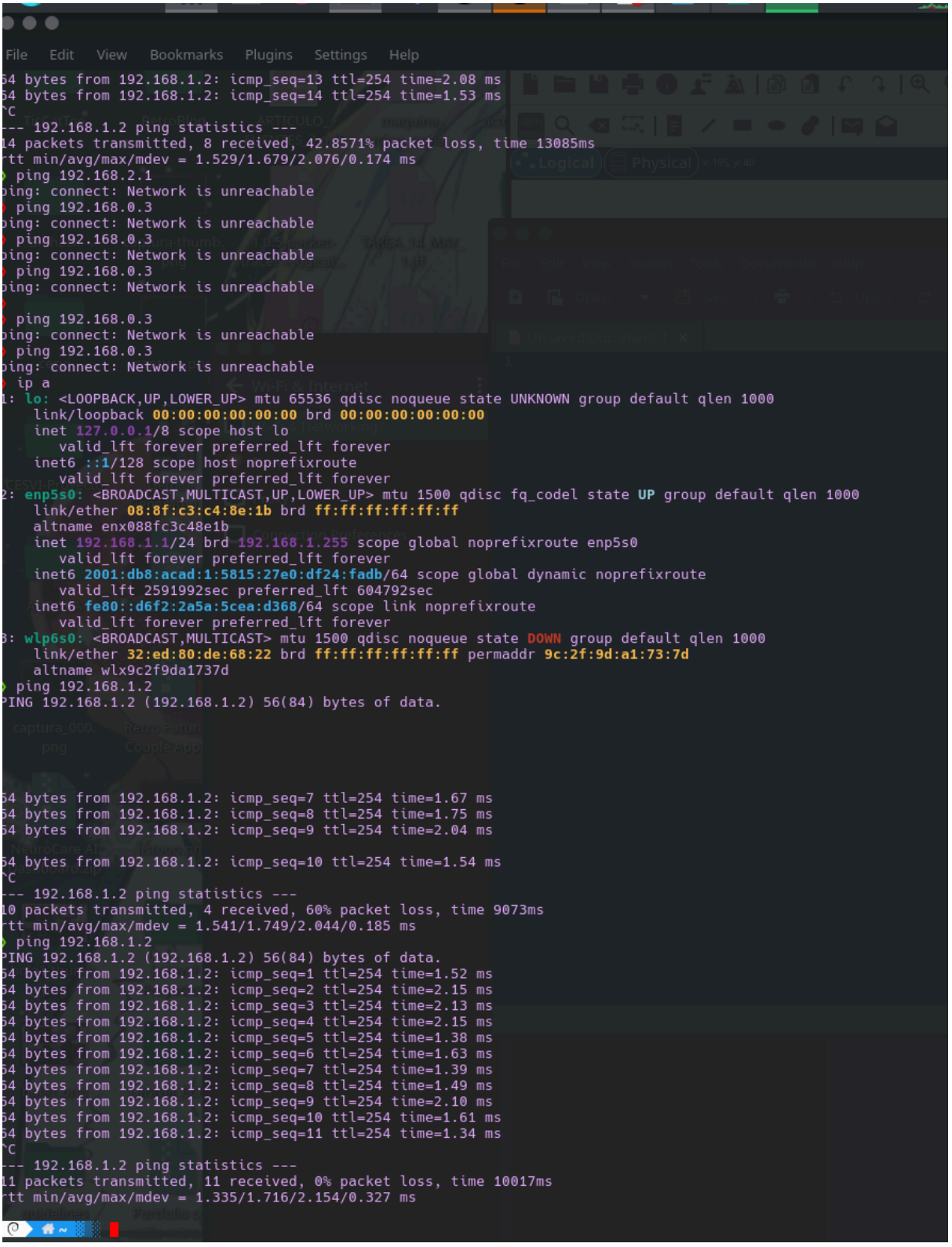
Porque los dispositivos intermedios aún no han sido configurados. Por lo que las interfaces están inactivas y no existe una ruta lógica para que puedan viajar los paquetes

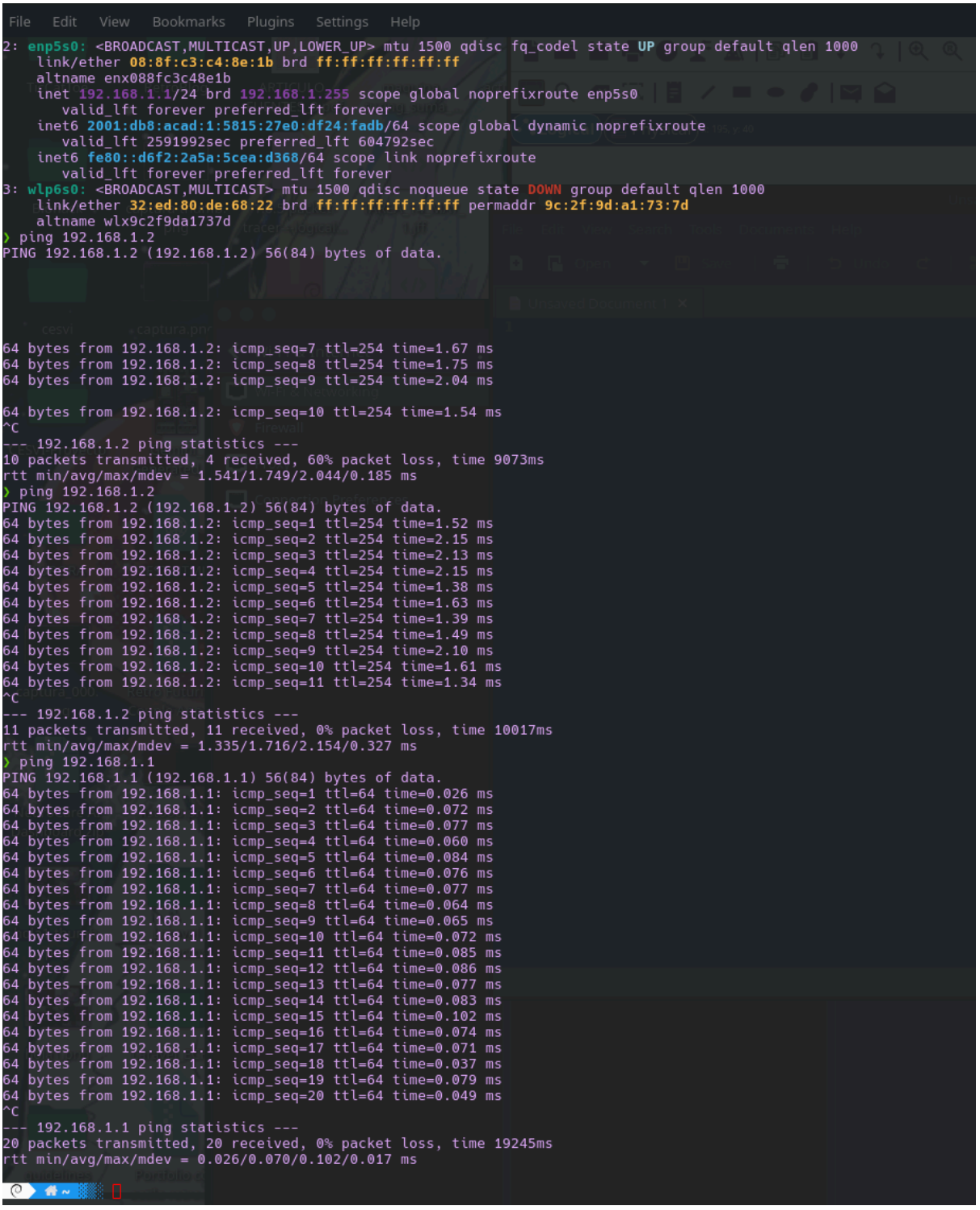
Después de la configuración del router:

¿Fueron correctos los pings?

En teoría fueron correctos, pero debido a la cuestión de abajo podemos ver que la comunicación solamente falla entre las pcs, pero el ping llega a todo.

Paso 4: **DEBIDO A LAS VERSIONES DE LOS SISTEMAS OPERATIVOS (EN ESTE CASO DEL ROUTER DE CISCO) NO ES POSIBLE LA COMUNICACIÓN ENTRE LA PC-A Y PC-B**


File Edit View Bookmarks Plugins Settings Help
64 bytes from 192.168.1.2: icmp_seq=13 ttl=254 time=2.08 ms
64 bytes from 192.168.1.2: icmp_seq=14 ttl=254 time=1.53 ms
^C
--- 192.168.1.2 ping statistics ---
14 packets transmitted, 8 received, 42.8571% packet loss, time 13085ms
rtt min/avg/max/mdev = 1.529/1.679/2.076/0.174 ms
ping 192.168.2.1
ping: connect: Network is unreachable
ping 192.168.0.3
ping: connect: Network is unreachable
ping 192.168.0.3
ping: connect: Network is unreachable
ping 192.168.0.3
ping: connect: Network is unreachable
ping 192.168.0.3
ping: connect: Network is unreachable
ping 192.168.0.3
ping: connect: Network is unreachable
ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
inet 127.0.0.1/8 scope host lo
valid_lft forever preferred_lft forever
inet6 ::1/128 scope host noprefixroute
valid_lft forever preferred_lft forever
2: enp5s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:8f:c3:c4:8e:1b brd ff:ff:ff:ff:ff:ff
altname enx088fc3c48e1b
inet 192.168.1.1/24 brd 192.168.1.255 scope global noprefixroute enp5s0
valid_lft forever preferred_lft forever
inet6 2001:db8:acad:1:5815:27e0:df24:fadb/64 scope global dynamic noprefixroute
valid_lft 2591992sec preferred_lft 604792sec
inet6 fe80::d6f2:2a5a:5cea:d368/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: wlp6s0: <BROADCAST,MULTICAST> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
link/ether 32:ed:80:de:68:22 brd ff:ff:ff:ff:ff:ff permaddr 9c:2f:9d:a1:73:7d
altname wlx9c2f9da1737d
ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=7 ttl=254 time=1.67 ms
64 bytes from 192.168.1.2: icmp_seq=8 ttl=254 time=1.75 ms
64 bytes from 192.168.1.2: icmp_seq=9 ttl=254 time=2.04 ms
64 bytes from 192.168.1.2: icmp_seq=10 ttl=254 time=1.54 ms
^C
--- 192.168.1.2 ping statistics ---
10 packets transmitted, 4 received, 60% packet loss, time 9073ms
rtt min/avg/max/mdev = 1.541/1.749/2.044/0.185 ms
ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=254 time=1.52 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=254 time=2.15 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=254 time=2.13 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=254 time=2.15 ms
64 bytes from 192.168.1.2: icmp_seq=5 ttl=254 time=1.38 ms
64 bytes from 192.168.1.2: icmp_seq=6 ttl=254 time=1.63 ms
64 bytes from 192.168.1.2: icmp_seq=7 ttl=254 time=1.39 ms
64 bytes from 192.168.1.2: icmp_seq=8 ttl=254 time=1.49 ms
64 bytes from 192.168.1.2: icmp_seq=9 ttl=254 time=2.10 ms
64 bytes from 192.168.1.2: icmp_seq=10 ttl=254 time=1.61 ms
64 bytes from 192.168.1.2: icmp_seq=11 ttl=254 time=1.34 ms
^C
--- 192.168.1.2 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10017ms
rtt min/avg/max/mdev = 1.335/1.716/2.154/0.327 ms


File Edit View Bookmarks Plugins Settings Help
2: enp5s0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
link/ether 08:8f:c3:c4:8e:1b brd ff:ff:ff:ff:ff:ff
altname enx088fc3c48e1b
inet 192.168.1.1/24 brd 192.168.1.255 scope global noprefixroute enp5s0
valid_lft forever preferred_lft forever
inet6 2001:db8:acad:1:5815:27e0:df24:fadb/64 scope global dynamic noprefixroute
valid_lft 2591992sec preferred_lft 604792sec
inet6 fe80::d6f2:2a5a:5cea:d368/64 scope link noprefixroute
valid_lft forever preferred_lft forever
3: wlp6s0: <BROADCAST,MULTICAST> mtu 1500 qdisc noqueue state DOWN group default qlen 1000
link/ether 32:ed:80:de:68:22 brd ff:ff:ff:ff:ff:ff permaddr 9c:2f:9d:a1:73:7d
altname wlx9c2f9da1737d
❯ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=7 ttl=254 time=1.67 ms
64 bytes from 192.168.1.2: icmp_seq=8 ttl=254 time=1.75 ms
64 bytes from 192.168.1.2: icmp_seq=9 ttl=254 time=2.04 ms
64 bytes from 192.168.1.2: icmp_seq=10 ttl=254 time=1.54 ms
^C
--- 192.168.1.2 ping statistics ---
10 packets transmitted, 4 received, 60% packet loss, time 9073ms
rtt min/avg/max/mdev = 1.541/1.749/2.044/0.185 ms
❯ ping 192.168.1.2
PING 192.168.1.2 (192.168.1.2) 56(84) bytes of data.
64 bytes from 192.168.1.2: icmp_seq=1 ttl=254 time=1.52 ms
64 bytes from 192.168.1.2: icmp_seq=2 ttl=254 time=2.15 ms
64 bytes from 192.168.1.2: icmp_seq=3 ttl=254 time=2.13 ms
64 bytes from 192.168.1.2: icmp_seq=4 ttl=254 time=2.15 ms
64 bytes from 192.168.1.2: icmp_seq=5 ttl=254 time=1.38 ms
64 bytes from 192.168.1.2: icmp_seq=6 ttl=254 time=1.63 ms
64 bytes from 192.168.1.2: icmp_seq=7 ttl=254 time=1.39 ms
64 bytes from 192.168.1.2: icmp_seq=8 ttl=254 time=1.49 ms
64 bytes from 192.168.1.2: icmp_seq=9 ttl=254 time=2.10 ms
64 bytes from 192.168.1.2: icmp_seq=10 ttl=254 time=1.61 ms
64 bytes from 192.168.1.2: icmp_seq=11 ttl=254 time=1.34 ms
^C
--- 192.168.1.2 ping statistics ---
11 packets transmitted, 11 received, 0% packet loss, time 10017ms
rtt min/avg/max/mdev = 1.335/1.716/2.154/0.327 ms
❯ ping 192.168.1.1
PING 192.168.1.1 (192.168.1.1) 56(84) bytes of data.
64 bytes from 192.168.1.1: icmp_seq=1 ttl=64 time=0.026 ms
64 bytes from 192.168.1.1: icmp_seq=2 ttl=64 time=0.072 ms
64 bytes from 192.168.1.1: icmp_seq=3 ttl=64 time=0.077 ms
64 bytes from 192.168.1.1: icmp_seq=4 ttl=64 time=0.060 ms
64 bytes from 192.168.1.1: icmp_seq=5 ttl=64 time=0.084 ms
64 bytes from 192.168.1.1: icmp_seq=6 ttl=64 time=0.076 ms
64 bytes from 192.168.1.1: icmp_seq=7 ttl=64 time=0.077 ms
64 bytes from 192.168.1.1: icmp_seq=8 ttl=64 time=0.064 ms
64 bytes from 192.168.1.1: icmp_seq=9 ttl=64 time=0.065 ms
64 bytes from 192.168.1.1: icmp_seq=10 ttl=64 time=0.072 ms
64 bytes from 192.168.1.1: icmp_seq=11 ttl=64 time=0.085 ms
64 bytes from 192.168.1.1: icmp_seq=12 ttl=64 time=0.086 ms
64 bytes from 192.168.1.1: icmp_seq=13 ttl=64 time=0.077 ms
64 bytes from 192.168.1.1: icmp_seq=14 ttl=64 time=0.083 ms
64 bytes from 192.168.1.1: icmp_seq=15 ttl=64 time=0.102 ms
64 bytes from 192.168.1.1: icmp_seq=16 ttl=64 time=0.074 ms
64 bytes from 192.168.1.1: icmp_seq=17 ttl=64 time=0.071 ms
64 bytes from 192.168.1.1: icmp_seq=18 ttl=64 time=0.037 ms
64 bytes from 192.168.1.1: icmp_seq=19 ttl=64 time=0.079 ms
64 bytes from 192.168.1.1: icmp_seq=20 ttl=64 time=0.049 ms
^C
--- 192.168.1.1 ping statistics ---
20 packets transmitted, 20 received, 0% packet loss, time 19245ms
rtt min/avg/max/mdev = 0.026/0.070/0.102/0.017 ms

```
C:\Users\chuom>ping 192.168.0.1

Haciendo ping a 192.168.0.1 con 32 bytes de datos:
Respuesta desde 192.168.0.1: bytes=32 tiempo=2ms TTL=255
Respuesta desde 192.168.0.1: bytes=32 tiempo=2ms TTL=255
Respuesta desde 192.168.0.1: bytes=32 tiempo=2ms TTL=255
Respuesta desde 192.168.0.1: bytes=32 tiempo=3ms TTL=255

Estadísticas de ping para 192.168.0.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 2ms, Máximo = 3ms, Media = 2ms

C:\Users\chuom>ping 192.168.1.1

Haciendo ping a 192.168.1.1 con 32 bytes de datos:
Respuesta desde 192.168.1.1: bytes=32 tiempo<1m TTL=255
Respuesta desde 192.168.1.1: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.1.1: bytes=32 tiempo=1ms TTL=255
Respuesta desde 192.168.1.1: bytes=32 tiempo=1ms TTL=255

Estadísticas de ping para 192.168.1.1:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 0ms, Máximo = 1ms, Media = 0ms
```

```
"192.168.1.3" no se reconoce como un comando interno o externo,
programa o archivo por lotes ejecutable.

C:\Users\chuom>ping 192.168.1.3

Haciendo ping a 192.168.1.3 con 32 bytes de datos:
Tiempo de espera agotado para esta solicitud.
Tiempo de espera agotado para esta solicitud.
Tiempo de espera agotado para esta solicitud.
Tiempo de espera agotado para esta solicitud.

Estadísticas de ping para 192.168.1.3:
    Paquetes: enviados = 4, recibidos = 0, perdidos = 4
    (100% perdidos),

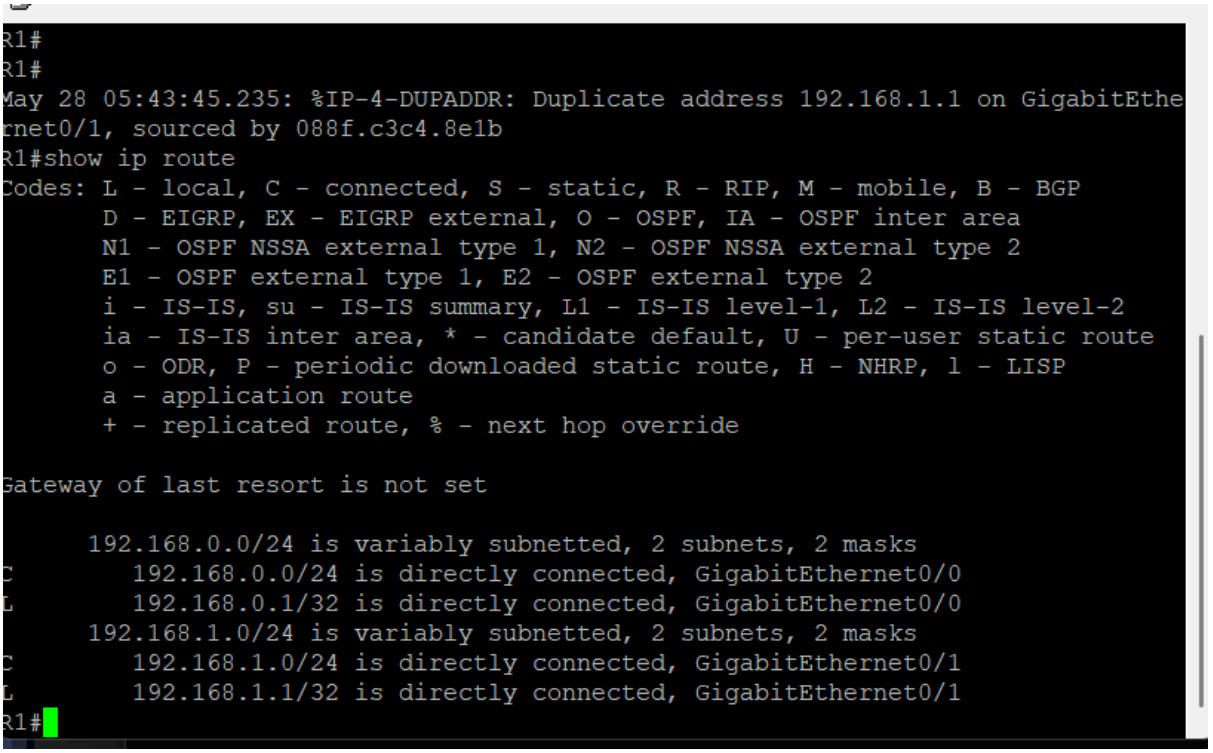
C:\Users\chuom>ping 192.168.1.2

Haciendo ping a 192.168.1.2 con 32 bytes de datos:
Respuesta desde 192.168.1.2: bytes=32 tiempo=3ms TTL=253
Respuesta desde 192.168.1.2: bytes=32 tiempo=2ms TTL=253
Respuesta desde 192.168.1.2: bytes=32 tiempo=3ms TTL=253
Respuesta desde 192.168.1.2: bytes=32 tiempo=2ms TTL=253

Estadísticas de ping para 192.168.1.2:
    Paquetes: enviados = 4, recibidos = 4, perdidos = 0
    (0% perdidos),
Tiempos aproximados de ida y vuelta en milisegundos:
    Mínimo = 2ms, Máximo = 3ms, Media = 2ms

C:\Users\chuom>
```

Parte 4, paso 1:



¿Qué código se utiliza en la tabla de routing para indicar una red conectada directamente?

"C"

¿Cuántas entradas de ruta están codificadas con un código C en la tabla de routing?

2

¿Qué tipos de interfaces están asociadas a las rutas con código C?

Las GigabitEthernet

Show ipv6 route:

```
C        192.168.0.0/24 is directly connected, GigabitEthernet0/0
L        192.168.0.1/32 is directly connected, GigabitEthernet0/0
     192.168.1.0/24 is variably subnetted, 2 subnets, 2 masks
C        192.168.1.0/24 is directly connected, GigabitEthernet0/1
L        192.168.1.1/32 is directly connected, GigabitEthernet0/1
R1#show ipv6 route
IPv6 Routing Table - default - 5 entries
Codes: C - Connected, L - Local, S - Static, U - Per-user Static route
       B - BGP, R - RIP, I1 - ISIS L1, I2 - ISIS L2
       IA - ISIS interarea, IS - ISIS summary, D - EIGRP, EX - EIGRP external
       ND - ND Default, NDp - ND Prefix, DCE - Destination, NDr - Redirect
       O - OSPF Intra, OI - OSPF Inter, OE1 - OSPF ext 1, OE2 - OSPF ext 2
       ON1 - OSPF NSSA ext 1, ON2 - OSPF NSSA ext 2, a - Application
C   2001:DB8:ACAD::/64 [0/0]
     via GigabitEthernet0/0, directly connected
L   2001:DB8:ACAD::1/128 [0/0]
     via GigabitEthernet0/0, receive
C   2001:DB8:ACAD:1::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
     via GigabitEthernet0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
R1#
```

Paso 2:

```
C   2001:DB8:ACAD:1::/64 [0/0]
     via GigabitEthernet0/1, directly connected
L   2001:DB8:ACAD:1::1/128 [0/0]
     via GigabitEthernet0/1, receive
L   FF00::/8 [0/0]
     via Null0, receive
R1#show interface g0/1
GigabitEthernet0/1 is up, line protocol is up
  Hardware is CN Gigabit Ethernet, address is 00c8.8b79.40a1 (bia 00c8.8b79.40a1
)
  Description: Conectado a S1
  Internet address is 192.168.1.1/24
  MTU 1500 bytes, BW 1000000 Kbit/sec, DLY 10 usec,
     reliability 255/255, txload 1/255, rxload 1/255
  Encapsulation ARPA, loopback not set
  Keepalive set (10 sec)
  Full Duplex, 1Gbps, media type is RJ45
  output flow-control is unsupported, input flow-control is unsupported
  ARP type: ARPA, ARP Timeout 04:00:00
  Last input 00:00:05, output 00:00:05, output hang never
  Last clearing of "show interface" counters never
  Input queue: 0/75/0/0 (size/max/drops/flushes); Total output drops: 0
  Queueing strategy: fifo
  Output queue: 0/40 (size/max)
  5 minute input rate 0 bits/sec, 0 packets/sec
  5 minute output rate 0 bits/sec, 0 packets/sec
     994 packets input, 144054 bytes, 0 no buffer
     Received 902 broadcasts (0 IP multicasts)
     0 runts, 0 giants, 0 throttles
     0 input errors, 0 CRC, 0 frame, 0 overrun, 0 ignored
     0 watchdog, 169 multicast, 0 pause input
     1060 packets output, 152045 bytes, 0 underruns
     0 output errors, 0 collisions, 1 interface resets
     94 unknown protocol drops
     0 babbles, 0 late collision, 0 deferred
     2 lost carrier, 0 no carrier, 0 pause output
     0 output buffer failures, 0 output buffers swapped out
R1#show interface g0/1
```

¿Cuál es el estado operativo de la interfaz G0/1?

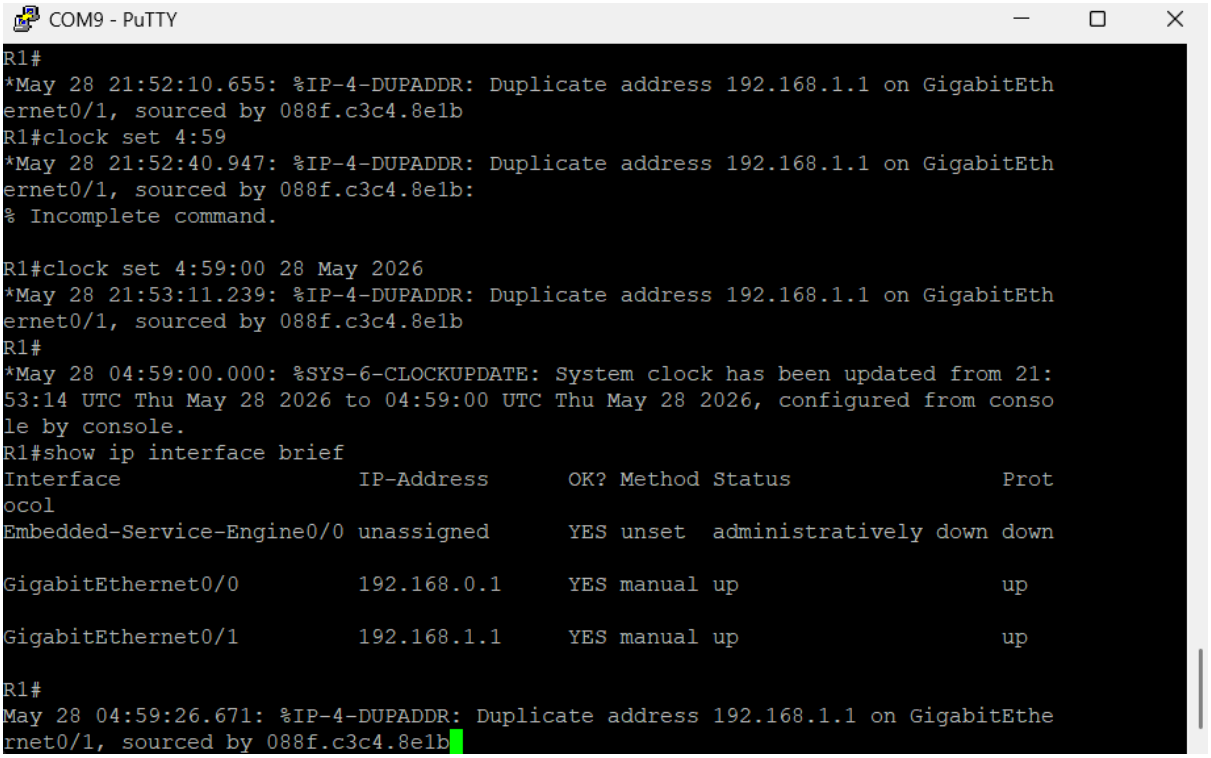
Como “is up”

¿Cuál es la dirección de control de acceso a los medios (MAC) de la interfaz G0/1?

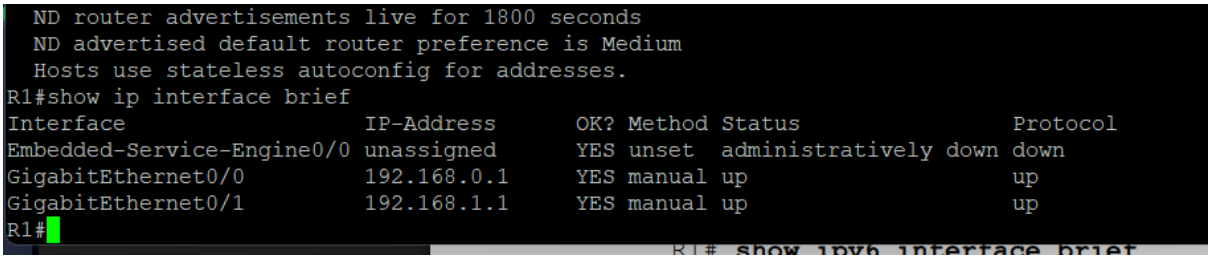
00c8.8b79.40a1.

¿Cómo se muestra la dirección de Internet en este comando?

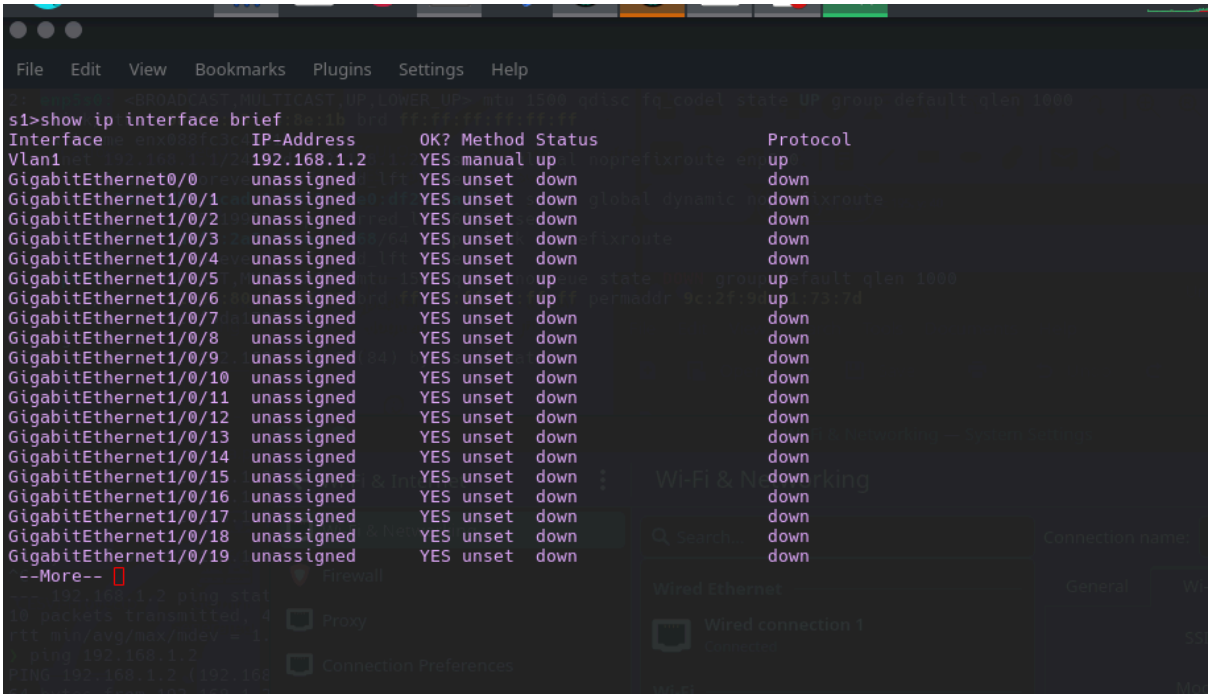
Como: Internet address is 192.168.1.1/24



Paso 3:

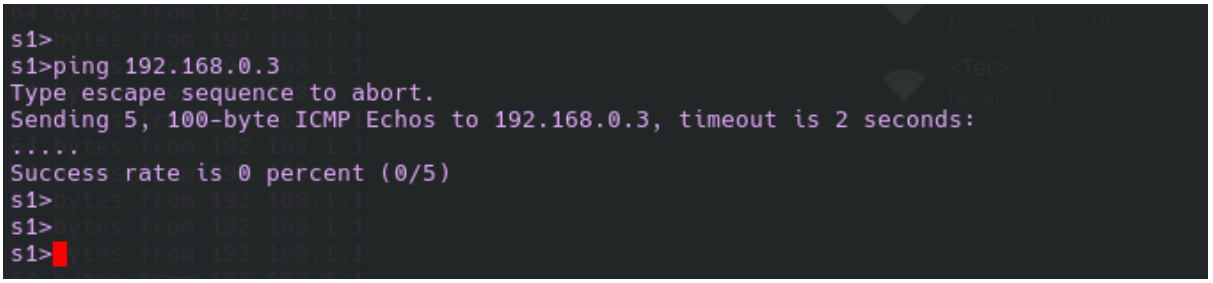


```
GigabitEthernet0/0         192.168.0.1     YES manual up
GigabitEthernet0/1         192.168.1.1     YES manual up
R1#show ipv6 interface brief
Em0/0                  [administratively down/down]
    unassigned
GigabitEthernet0/0     [up/up]
    FE80::1
    2001:DB8:ACAD::1
GigabitEthernet0/1     [up/up]
    FE80::1
    2001:DB8:ACAD:1::1
R1#
```



```
 ND advertised default router preference is Medium
 Hosts use stateless autoconfig for addresses.
R1#show ipv6 interface g0/0
GigabitEthernet0/0 is up, line protocol is up
 IPv6 is enabled, link-local address is FE80::1
 No Virtual link-local address(es):
 Description: Conectado a PC-B
 Global unicast address(es):
   2001:DB8:ACAD::1, subnet is 2001:DB8:ACAD::/64
 Joined group address(es):
   FF02::1
   FF02::2
   FF02::1:FF00:1
 MTU is 1500 bytes
 ICMP error messages limited to one every 100 milliseconds
 ICMP redirects are enabled
 ICMP unreachables are sent
 ND DAD is enabled, number of DAD attempts: 1
 ND reachable time is 30000 milliseconds (using 30000)
 ND advertised reachable time is 0 (unspecified)
 ND advertised retransmit interval is 0 (unspecified)
 ND router advertisements are sent every 200 seconds
 ND router advertisements live for 1800 seconds
 ND advertised default router preference is Medium
 Hosts use stateless autoconfig for addresses.
R1#
```

```
R1#show ipv6 interface g0/1
GigabitEthernet0/1 is up, line protocol is up
  IPv6 is enabled, link-local address is FE80::1
  No Virtual link-local address(es):
  Description: Conectado a S1
  Global unicast address(es):
    2001:DB8:ACAD:1::1, subnet is 2001:DB8:ACAD:1::/64
  Joined group address(es):
    FF02::1
    FF02::2
    FF02::1:FF00:1
  MTU is 1500 bytes
  ICMP error messages limited to one every 100 milliseconds
  ICMP redirects are enabled
  ICMP unreachables are sent
  ND DAD is enabled, number of DAD attempts: 1
  ND reachable time is 30000 milliseconds (using 30000)
  ND advertised reachable time is 0 (unspecified)
  ND advertised retransmit interval is 0 (unspecified)
  ND router advertisements are sent every 200 seconds
  ND router advertisements live for 1800 seconds
  ND advertised default router preference is Medium
  Hosts use stateless autoconfig for addresses.
R1#
```



Preguntas de reflexión:
Si la interfaz G0/1 se mostrara administrativamente inactiva, ¿qué comando de configuración de interfaz usaría para activar la interfaz?

Se debe entrar al modo de configuración de interfaz y ejecutar el “no shutdown”

¿Qué ocurriría si hubiera configurado incorrectamente la interfaz G0/1 en el router con una dirección IP
192.168.1.2?

No se podría hacer ping a la PC-B desde PC-A porque la dirección IP estaría asignada al S1. Al intentar asignar la misma IP a la interfaz del router, ocurriría un conflicto de direcciones IP, y el router no permitiría que la interfaz funcione correctamente en esa red, rompiendo el gateway predeterminado para los dispositivos de esa subred.